



2 Two binary numbers are stored in a computer.

(a) Add the **two** binary numbers using binary addition.

Give your answer in binary. Show all of your working.

$$\begin{array}{r} 11110101 \\ + 00111001 \\ \hline \end{array}$$

[4]

(b) A logic right shift of **three** places is performed on the binary number 11110101.

Give the denary number for the binary number that would be stored after the logical shift of **three** places has occurred. Show all of your working.

Working space

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Denary number [3]

3 A programmer uses assembly language to create a program for a coffee machine.

(a) Describe what is meant by assembly language.

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..... [2]

(b) Identify the translator needed for assembly language.

..... [1]

(c) Tick (✓) **one** box to show which statement is an advantage of using assembly language.

- | | |
|----------------------------------|--------------------------|
| A It is close to human language. | ☐ |
| B It is memory efficient. | ☐ |
| C It is easy to debug. | ☐ |
| D It is machine independent. | ☐ |

[1]

